35.

AIM:

Create a database in mysql and write a menu-based program to perform the operations of insertion, selection, updation and deletion through python cursor programs.

Program:

```
import MySQLdb
con = MySQLdb.connect(host="localhost", user="root", password="windows",
database="employee")
cur = con.cursor()

# st="create table emp(empno int(5) primary key, name varchar(25),salary int(6),design
varchar(20))"

while True:
    print("\n1.insert\n2.display\n3.update\n4.delete\n5.search")
    print("enter your choice:")
    ch = int(input("enter choice:"))

    if ch == 1:
        no = int(input("enter emp number: "))
        name = input(" enter emp name: ")
        sal = int(input("enter emp salary: "))
        desg = input("enter designation: ")
        st = "insert into emp values({},'{}',{},'{}')".format(no, name, sal, desg)
        cur.execute(st)
        print("inserted successfully!!")
        con.commit()

    elif ch == 2:
        st = "select * from emp"
        cur.execute(st)
        data = cur.fetchall()
        for row in data:
            print("empno={} name='{}' salary={} design='{}'".format(row[0], row[1], row[2], row[3]))

    elif ch == 3:
        no = int(input("enter empno to update: "))
        name = input("enter name to update: ")
        st = "update emp set name='{}' where empno={}".format(name, no)
        cur.execute(st)
        print("updated successfully")
        con.commit()

    elif ch == 4:
        no = int(input("enter empno to remove: "))
        st = "delete from emp where empno={}".format(no)
        cur.execute(st)
```

```
        print("deleted successfully!!")
        con.commit()

    elif ch == 5:
        no = int(input("enter emp number to search in employee: "))
        st = "select * from emp where empno={}".format(no)
        cur.execute(st)
        data = cur.fetchone()
        print("empno={} name='{}' sal={} desig='{}'".format(data[0], data[1], data[2], data[3]))

    elif ch == 6:
        break

con.close()
```

Output:

```
inserted successfully!!

1.insert
2.display
3.update
4.delete
5.search
enter your choice:
enter choice:1
enter emp number: 234
 enter emp name: Amrit
enter emp salary: 45000
enter designation: IT
inserted successfully!!

1.insert
2.display
3.update
4.delete
5.search
enter your choice:
enter choice:2
empno=101 name='Swaroop' salary=25000 design='Ac
empno=102 name='Sanjay' salary=30000 design='IT'
empno=234 name='Amrit' salary=45000 design='IT'

1.insert
2.display
3.update
4.delete
5.search
enter your choice:
enter choice:3
enter empno to update: 102
enter name to update: Manoj
updated successfully
```

```
1.insert
2.display
3.update
4.delete
5.search
enter your choice:
enter choice:4
enter empno to remove: 234
deleted successfully!!
```

```
1.insert
2.display
3.update
4.delete
5.search
enter your choice:
enter choice:5
enter emp number to search in employee: 101
empno=101 name='Swaroop' sal=25000 desig='Accounts'

1.insert
2.display
3.update
4.delete
5.search
enter your choice:
enter choice:
```

36.

AIM:

Write a program to

(i) Print current date and time in string format.

(ii) Find square root of a number.

Program:

```
from math import*
from datetime import datetime
x=float(input("enter a number"))
sq=sqrt(x)
print("Square root of {} is {}".format(x,sq))

now=datetime.now()
dt=now.strftime("%d/%m/%Y")
tm=now.strftime("%H:%M:%S")
print("date=",dt)
print("time=",tm)
```

Output:

```
sanjay@sanjay:~/Desktop/IMCA440$ python module2.py
enter a number100
Square root of 100.0 is 10.0
date= 11/02/2025
time= 08:32:40
```

37.

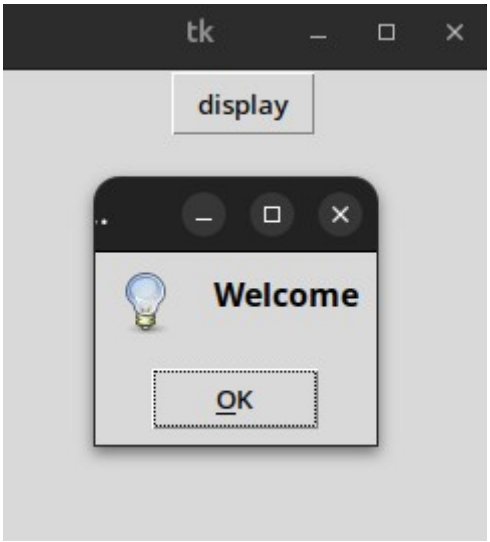
AIM:

Write a tkinter program to display a message box.

Program:

```
from tkinter import *
from tkinter import messagebox
root=Tk()
root.geometry("200x200")
def disp():
	messagebox.showinfo("IMCAB9","Welcome")
B=Button(root,text="display",command=disp)
B.pack()
root.mainloop()
```

Output:



38.

AIM:

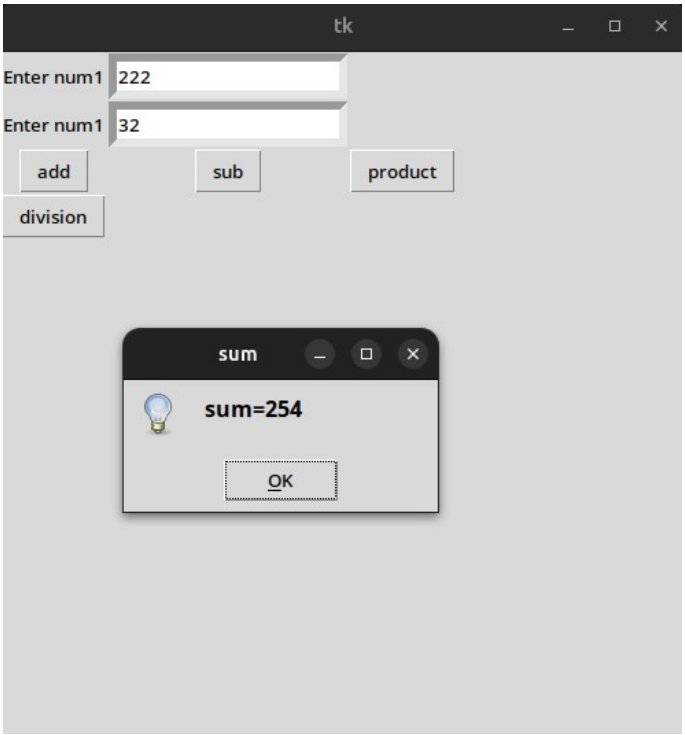
Write a tkinter program to implement a calculator

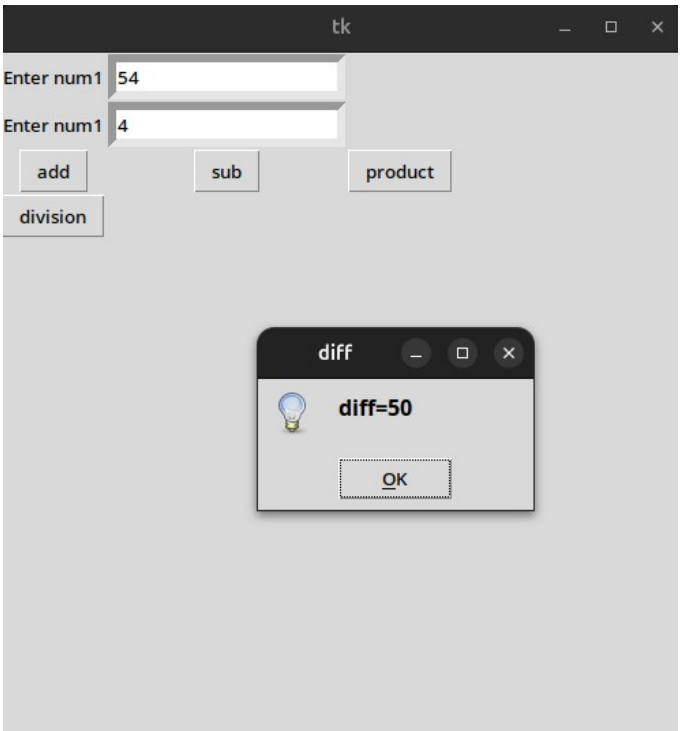
Program:

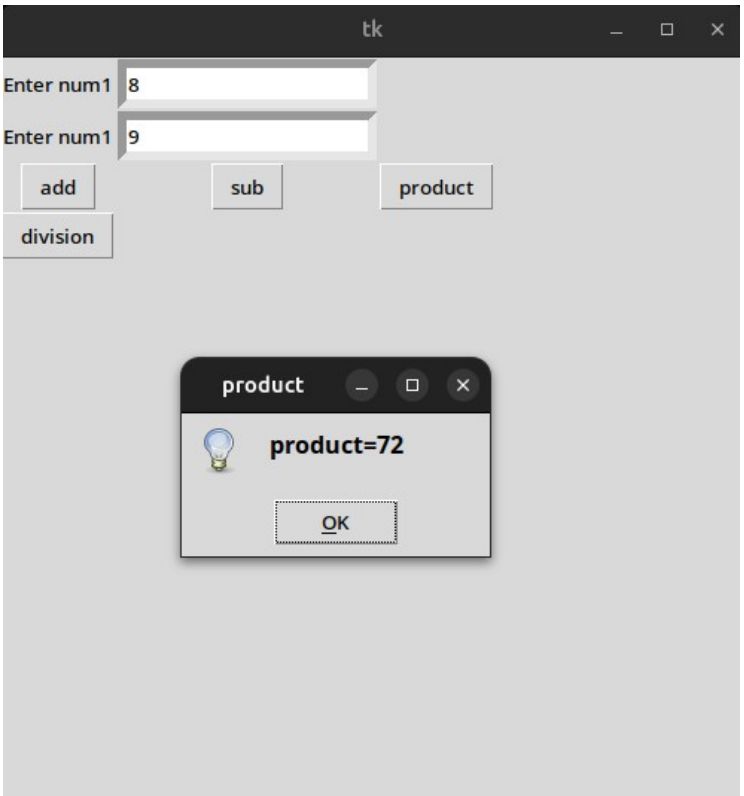
```
from tkinter import*
from tkinter import messagebox
top=Tk()
top.geometry("500x500")
L1=Label(top,text="Enter num1")
L1.grid(row=1,column=1)
L2=Label(top,text="Enter num1")
L2.grid(row=2,column=1)
E1=Entry(top,bd=6)
E1.grid(row=1,column=2)
E2=Entry(top,bd=6)
E2.grid(row=2,column=2)
def sum():
	n1=int(E1.get())
	n2=int(E2.get())
	x=n1+n2
	messagebox.showinfo("sum","sum={}".format(x))
	E1.delete(0,END)
	E2.delete(0,END)
B1=Button(top,text="add",command=sum)

def sub():
	n1=int(E1.get())
	n2=int(E2.get())
	x=n1-n2
	messagebox.showinfo("diff","diff={}".format(x))
	E1.delete(0,END)
	E2.delete(0,END)
B2=Button(top,text="sub",command=sub)

def prod():
	n1=int(E1.get())
	n2=int(E2.get())
	x=n1*n2
	messagebox.showinfo("product","product={}".format(x))
	E1.delete(0,END)
	E2.delete(0,END)
B3=Button(top,text="product",command=prod)

def div():
	n1=int(E1.get())
	n2=int(E2.get())
```

```
        x=n1/n2
        messagebox.showinfo("division","division={}".format(x))
        E1.delete(0,END)
        E2.delete(0,END)
B4=Button(top,text="division",command=div)

B1.grid(row=3,column=1)
B2.grid(row=3,column=2)
B3.grid(row=3,column=3)
B4.grid(row=4,column=1)
top.mainloop()
```

Output:


tk
Enter num1
222
Enter num1
32
add
sub
product
division
sum
sum=254
OK


tk
Enter num1
54
Enter num1
4
add
sub
product
division
diff
diff=50
OK


tk
Enter num1
8
Enter num1
9
add
sub
product
division
product
product=72
OK

tk
Enter num1
90
Enter num1
8
add
sub
product
division
divi...
division=11.25
OK

39.

AIM:

Create a static webpage(3 pages) for an organization (educational or business) with images.

Program:

<u>Settings.py</u>

```
"""
Django settings for organization_site project.
Generated by 'django-admin startproject' using Django 5.1.6.
For more information on this file, see
https://docs.djangoproject.com/en/5.1/topics/settings/
For the full list of settings and their values, see
https://docs.djangoproject.com/en/5.1/ref/settings/
"""
from pathlib import Path
BASE_DIR = Path(__file__).resolve().parent.parent
SECRET_KEY = 'django-insecure-em*_^9(sbv=y%uwah$#3ju1geb2&uy^o5uq8f3dgscc&#wn=i!'
DEBUG = True
ALLOWED_HOSTS = []
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',

   'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]
MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
```

```
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
]
ROOT_URLCONF = 'organization_site.urls'
from pathlib import Path
BASE_DIR = Path(__file__).resolve().parent.parent
import os
TEMPLATES = [
    {
        'BACKEND': 'django.template.backends.django.DjangoTemplates',
        'DIRS': [os.path.join(BASE_DIR, 'pages', 'templates')],  # Points to pages/templates
        'APP_DIRS': True,



'OPTIONS': {
            'context_processors': [
                'django.template.context_processors.debug',
                'django.template.context_processors.request',
                'django.contrib.auth.context_processors.auth',
                'django.contrib.messages.context_processors.messages',
            ],
        },
    },
]
WSGI_APPLICATION = 'organization_site.wsgi.application'
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',
        'NAME': BASE_DIR / 'db.sqlite3',
    }
}
```

```
AUTH_PASSWORD_VALIDATORS = [
    {
        'NAME': 'django.contrib.auth.password_validation.UserAttributeSimilarityValidator',
    },
    {



      'NAME': 'django.contrib.auth.password_validation.MinimumLengthValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.CommonPasswordValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.NumericPasswordValidator',
    },
]
LANGUAGE_CODE = 'en-us'
TIME_ZONE = 'UTC'
USE_I18N = True
USE_TZ = True
STATIC_URL = '/static/'
STATICFILES_DIRS = [
    os.path.join(BASE_DIR, 'pages', 'static'),  # Adjusted to point to pages/static
]
DEFAULT_AUTO_FIELD = 'django.db.models.BigAutoField'
```

Urls.py

```
from django.contrib import admin
from django.urls import path
from pages import views
urlpatterns = [
    path('admin/', admin.site.urls),
```

```
    path('home/', views.home, name='home'),
    path('about/', views.about, name='about'),
    path('contact/', views.contact, name='contact'),
]
```

Views.py

```
from django.shortcuts import render
def home(request):
    return render(request, 'home.html')
def about(request):
    return render(request, 'about.html')
def contact(request):
    return render(request, 'contact.html')
```

home.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Home Page</title>
    {% load static %}
</head>
<body>
    <h1>Welcome to Our Organization</h1>
<img src="{% static 'images/download.png' %}" alt="Logo">
    <nav>
        <a href="/">Home</a>
        <a href="/about/">About</a>
        <a href="/contact/">Contact</a>
    </nav>
    <p>This is the homepage of our organization. We provide high-quality educational
resources.</p>
</body>
```

```
</html>
```

About.html

```
<!DOCTYPE html>
<html><head>
    <title>About Us</title>
    {% load static %}
</head>
<body> <h1>About Us</h1>
  <img src="{% static 'images/download.png' %}" alt="Logo">
    <p>Welcome to the About page!</p>
</body></html>
```

Contact.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Contact Us</title>
    {% load static %}
</head>
<body>
    <h1>Contact Us</h1>
     <img src="{% static 'images/download.png' %}" alt="Logo">
    <p>Feel free to reach out to us!</p>
</body>
</html>
```

Output:

Home.html

Welcome to Our Organization

Home About Contact

This is the homepage of our organization. We provide high-quality educational resources.

About.html

About Us

Welcome to the About page!

contact.html

Contact Us

Feel free to reach out to us!

40.

AIM:

Dynamic web page creation with GUI Interface.

Program:

forms.py

```
from django.test import TestCase
from django import forms
class StudForm(forms.Form):
name=forms.CharField(label="enter name",max_length=30)
m1=forms.DecimalField(label="enter math marks")
m2=forms.DecimalField(label="entetr science marks")
```

Views.py

```
from django.shortcuts import render
from django.http import HttpResponse
from .forms import StudForm
def welcome(request):
#return HttpResponse("HELLO WORLD")
return render(request,'home.html')
def calc(request):
if request.method == 'POST':
form=StudForm(request.POST)
if form.is_valid():
name=form.cleaned_data["name"]
mark1=form.cleaned_data["m1"]
mark2=form.cleaned_data["m2"]
tot=mark1+mark2
return HttpResponse('total marks for {} is {}'.format(name,tot))
else:
form=StudForm()
context={'form':form}
return render(request,'marks.html',context)

# Create your views here.
```

Urls.py

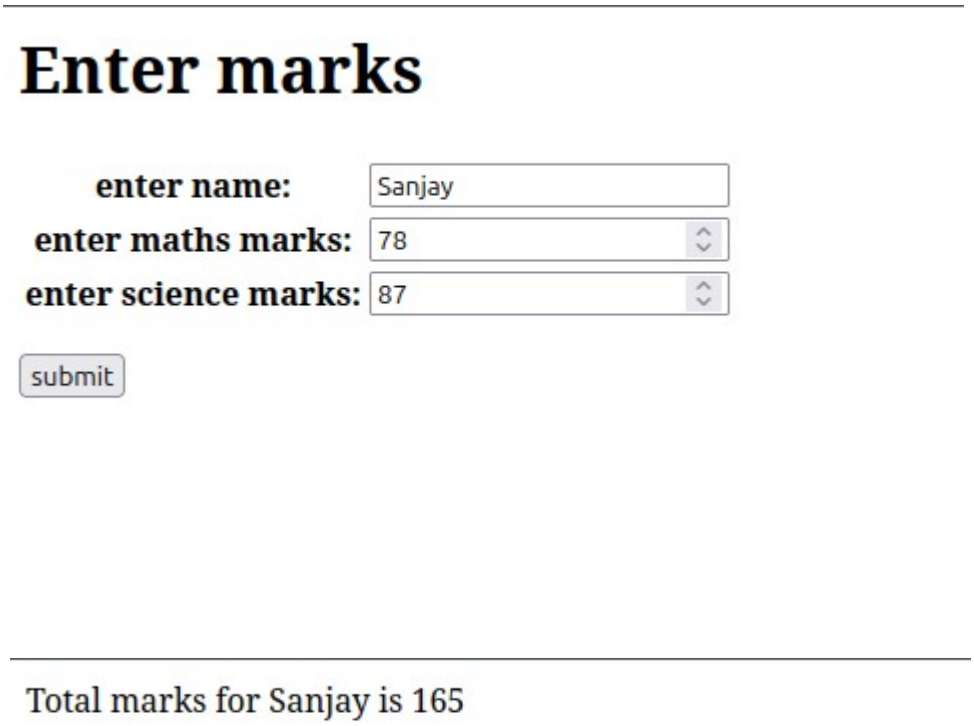
```
from django.urls import path
from . import views

urlpatterns = [
    path('wel/',views.welcome,name='views.welcome'),
```

```
    path('marks/',views.calc,name='views.calc'),
]
```

marks.html

```
<html>
<title>MARKS PAGE</title>
<body>
<h1> enter Marks</h1>
<form action= "." method="POST">
{% csrf_token %}
<table>
{{form.as_table}}
</table>
<p><input type="submit" value="submit"></p>
</forms>
</body>
</html>
```

Output:


Enter marks
enter name:
Sanjay
enter maths marks:
78
enter science marks:
87
submit
Total marks for Sanjay is 165